

# Perfect Competition

Based on Ch. 12, Economics 10th ed., Parkin, Powell & Mathews

Course: Eco 101 (Introduction to Microeconomics)

# Perfect Competition

The theory of **perfect competition** is built on **four** assumptions:

1. Many firms sell identical products to many buyers
1. There are no restrictions on entry into the market
1. Established firms have no advantage over new ones
1. Buyers and sellers are well informed about prices

# A Perfectly Competitive Firm Is a Price Taker

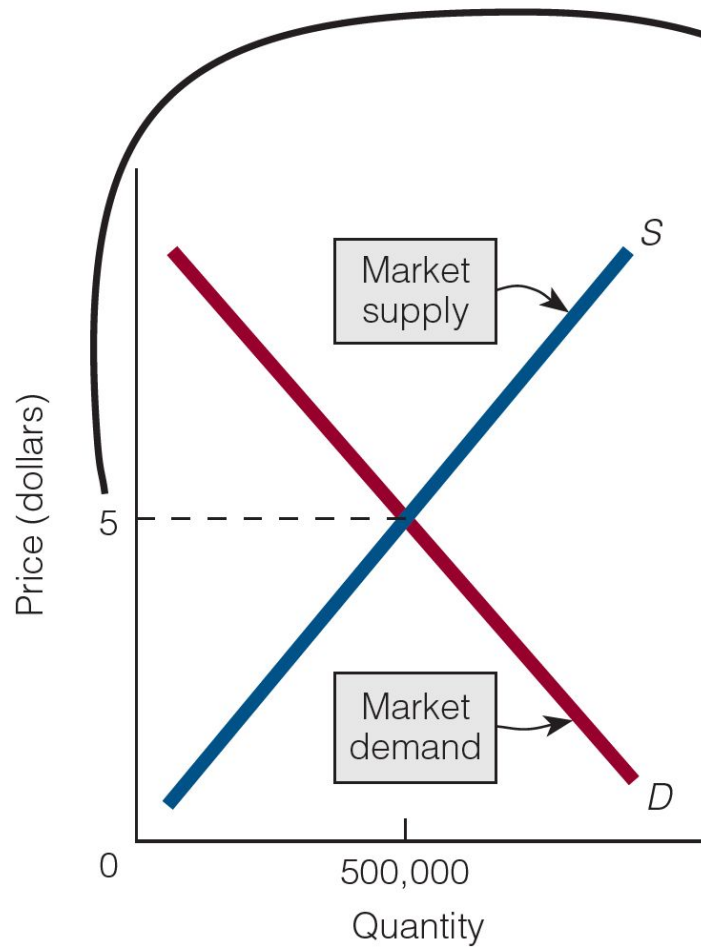
**Price Taker** – A seller that does not have the ability to control the price of the product it sells; the seller “takes” the price determined in the market.

# **Why Does a Perfectly Competitive Firm Sell at the Equilibrium Price?**

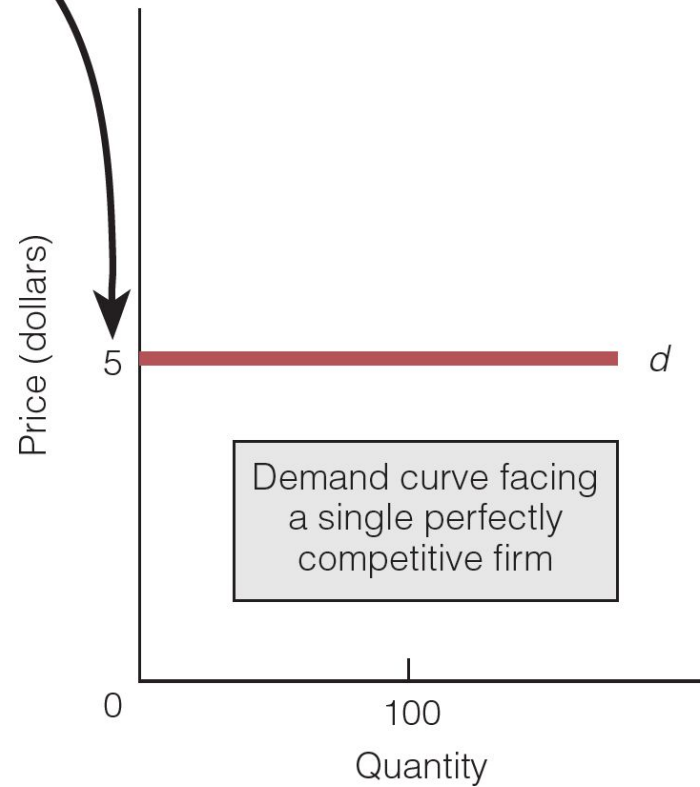
If a perfectly competitive firm charges a price higher than the market price, it won't sell any products. This is because the firm sells a product that is the same as others, its supply is small compared to the total market, and buyers know where to find the product at a lower price.

# The Market Demand Curve and Firm Demand Curve in Perfect Competition

- The market, composed of all buyers and sellers, establishes the equilibrium price
- **A single perfectly competitive firm then faces a horizontal (flat, perfectly elastic) demand curve**
- We conclude that the firm is a **price taker**: it takes the equilibrium price established by the market and sells any and all quantities of output at this price.



**(a) Market**



**(b) Single Firm**

# The Revenue of a Competitive Firm

A firm's goal is to **maximize economic profit**

$$\text{Economic profit} = \text{Total revenue} - \text{Total cost}$$

**Total revenue:** Total Revenue (TR) = **Price × Quantity sold**

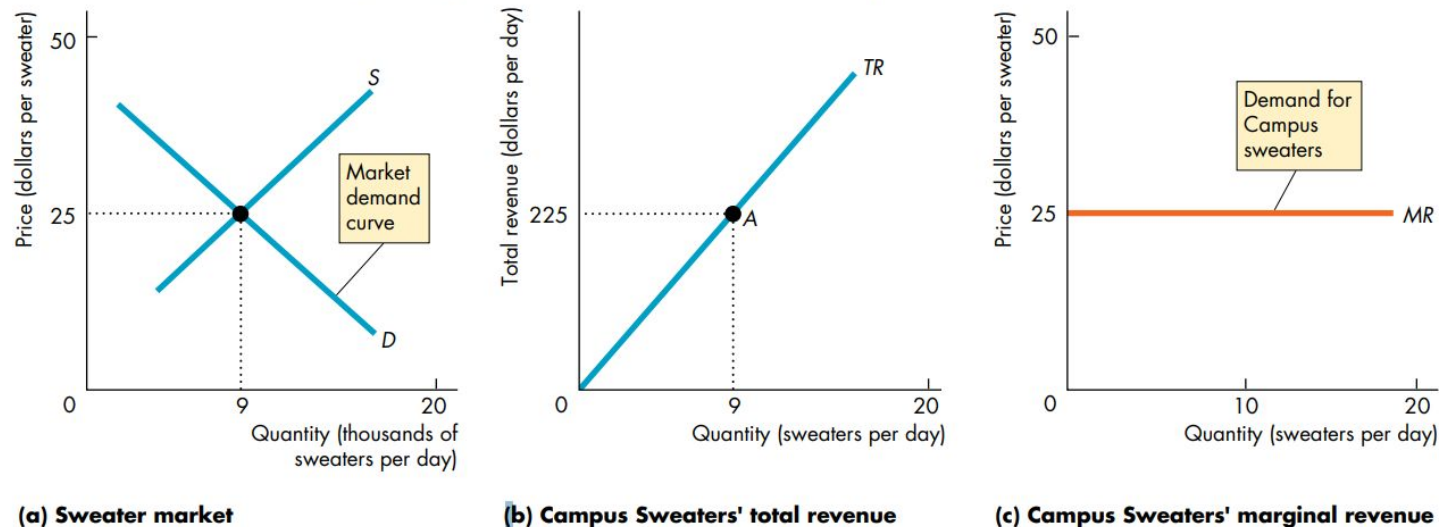
Example: Selling 9 sweaters at \$25 each → **TR = \$225**

**Marginal revenue:** Marginal revenue is the change in total revenue that results from a one-unit increase in quantity sold.

$$MR = \frac{\Delta TR}{\Delta Q}$$

Example: TR increases from \$200 to \$225 when quantity increases from 8 to 9 → **MR = \$25**

**FIGURE 12.1** Demand, Price, and Revenue in Perfect Competition



| Quantity sold<br>(Q)<br>(sweaters<br>per day) | Price<br>(P)<br>(dollars<br>per<br>sweater) | Total<br>revenue<br>( $TR = P \times Q$ )<br>(dollars) | Marginal<br>revenue<br>( $MR = \Delta TR / \Delta Q$ )<br>(dollars per<br>additional sweater) |
|-----------------------------------------------|---------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 8                                             | 25                                          | 200                                                    | 25                                                                                            |
| 9                                             | 25                                          | 225                                                    | 25                                                                                            |
| 10                                            | 25                                          | 250                                                    |                                                                                               |

In part (a), market demand and market supply determine the market price (and quantity). Part (b) shows the firm's total revenue curve ( $TR$ ). Point A corresponds to the second row of the table—Campus Sweaters sells 9 sweaters at \$25 a sweater, so total revenue is \$225. Part (c) shows the firm's marginal revenue curve ( $MR$ ). This curve is also the demand curve for the firm's sweaters. The demand for sweaters from Campus Sweaters is perfectly elastic at the market price of \$25 a sweater.



# Continued

- For a perfectly competitive firm, therefore, price (P) is equal to marginal revenue:

For a perfectly competitive firm,  $P = MR$

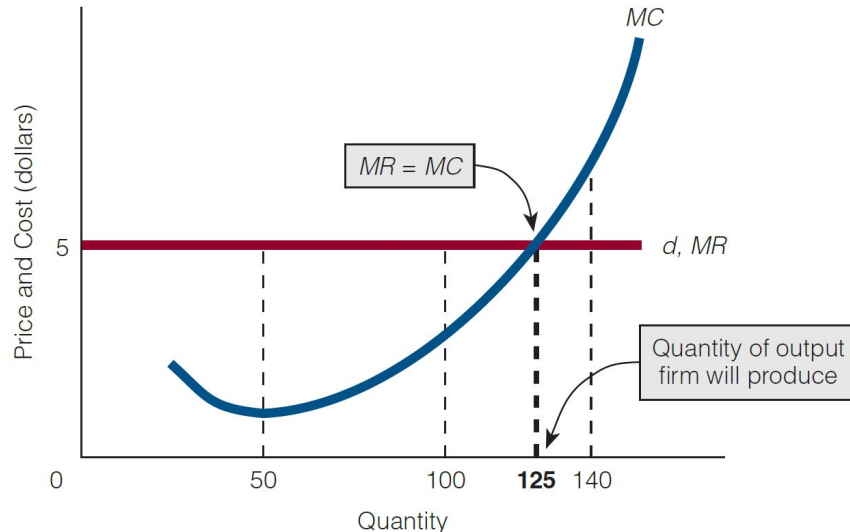
- As price is equal to marginal revenue, then the marginal revenue curve for the perfectly competitive firm is the same as its demand curve.

For a perfectly competitive firm, Demand curve = Marginal revenue curve

# Profit Maximization for a Competitive Firm

## The Quantity of Output That the Perfectly Competitive Firm Will Produce

The firm's demand curve is horizontal at the equilibrium price. Its demand curve is its marginal revenue curve. The firm produces that quantity of output at which  $MR = MC$ .



- If marginal revenue exceeds marginal cost, the firm should increase its output.
- If marginal cost exceeds marginal revenue, the firm should decrease its output.
- **At the profit-maximizing level of output, marginal revenue equals marginal cost**
- **Profit is maximized when  $P = MR = MC$**

# The Firm's Short-Run Decision to Shut Down

A **shutdown** is a **temporary decision** not to produce due to poor market conditions.

In a shutdown:

- **Output (Q) = 0**
- **Variable costs = 0**
- **Revenue = 0**
- But **fixed costs are still paid**

Therefore:

**Economic loss = Total Fixed Cost (TFC)**

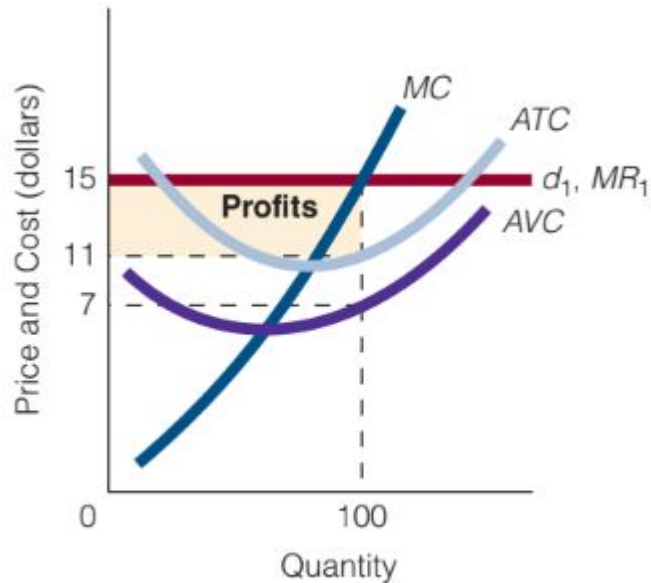
## When Does a Firm Shut Down?

- The **shutdown point** is when the firm is **indifferent** between producing and not producing.
- Occurs at the **minimum point on the AVC curve**
- Rule:
  - **Shut down if Price < Minimum AVC**
  - **Stay open if Price  $\geq$  Minimum AVC**

If the **price is exactly equal to the minimum AVC**, the firm is **indifferent between shutting down and continuing production** at that output level.

# To Produce or NOT to Produce?

## Case 1: Price is Above Average Total Cost

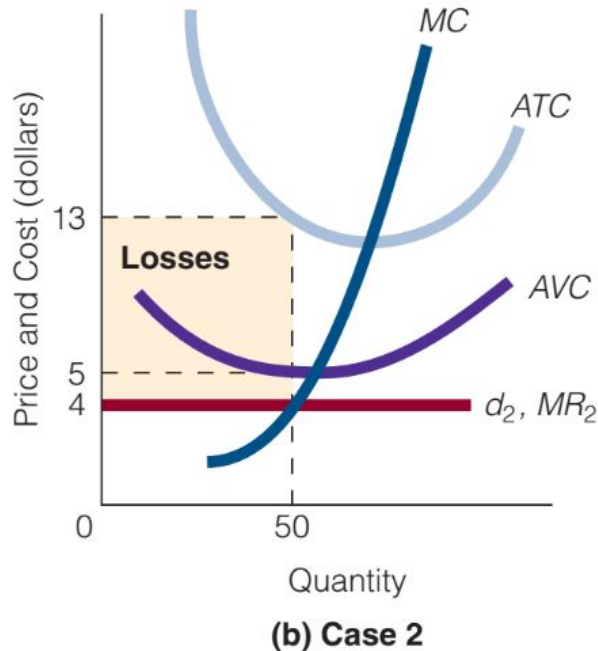


(a) Case 1

| Case 1                                                  |             |
|---------------------------------------------------------|-------------|
| Equilibrium price ( $P$ )                               | = \$15      |
| Quantity of output produced ( $Q$ )                     | = 100 units |
| Total revenue ( $P \times Q = \$15 \times 100$ )        | = \$1,500   |
| Total cost ( $ATC \times Q = \$11 \times 100$ )         | = \$1,100   |
| Total variable cost ( $AVC \times Q = \$7 \times 100$ ) | = \$700     |
| Total fixed cost ( $TC - TVC = \$1,100 - \$700$ )       | = \$400     |
| Profits ( $TR - TC = \$1,500 - \$1,100$ )               | = \$400     |

# To Produce or NOT to Produce? Cont.

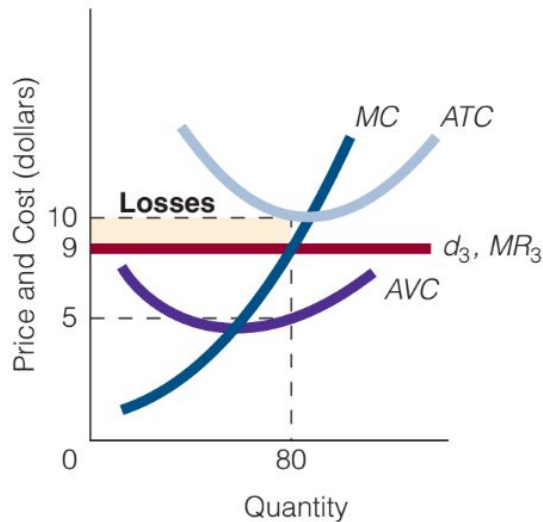
## Case 2: Price Is Below Average Variable Cost



| Case 2                                                 |            |
|--------------------------------------------------------|------------|
| Equilibrium price ( $P$ )                              | = \$4      |
| Quantity of output produced ( $Q$ )                    | = 50 units |
| Total revenue ( $P \times Q = \$4 \times 50$ )         | = \$200    |
| Total cost ( $ATC \times Q = \$13 \times 50$ )         | = \$650    |
| Total variable cost ( $AVC \times Q = \$5 \times 50$ ) | = \$250    |
| Total fixed cost ( $TC - TVC = \$650 - \$250$ )        | = \$400    |
| Profits ( $TR - TC = \$200 - \$650$ )                  | = -\$450   |

# To Produce or NOT to Produce? Cont.

## Case 3: Price Is Below Average Total Cost but Above Average Variable Cost



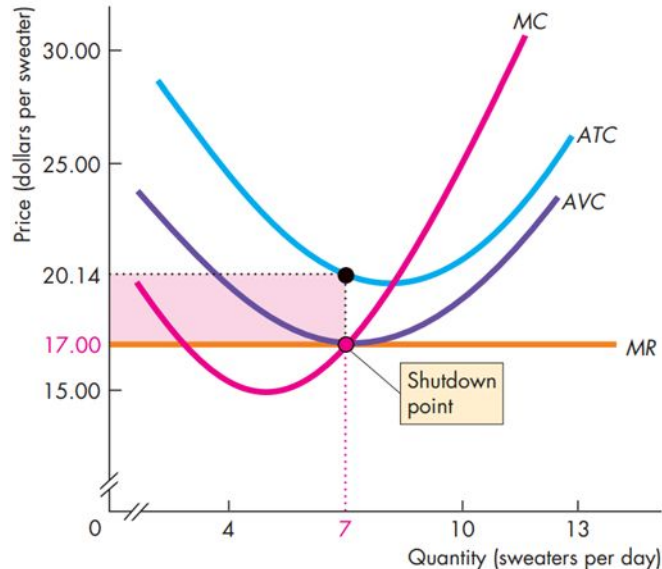
(c) Case 3

| Case 3                                                 |            |
|--------------------------------------------------------|------------|
| Equilibrium price ( $P$ )                              | = \$9      |
| Quantity of output produced ( $Q$ )                    | = 80 units |
| Total revenue ( $P \times Q = \$9 \times 80$ )         | = \$720    |
| Total cost ( $ATC \times Q = \$10 \times 80$ )         | = \$800    |
| Total variable cost ( $AVC \times Q = \$5 \times 80$ ) | = \$400    |
| Total fixed cost ( $TC - TVC = \$800 - \$400$ )        | = \$400    |
| Profits ( $TR - TC = \$720 - \$800$ )                  | = -\$80    |

# To Produce or NOT to Produce? Cont.

## Case 4: Price Is Equal to Average Variable Cost

FIGURE 12.4 The Shutdown Decision



### Case 4

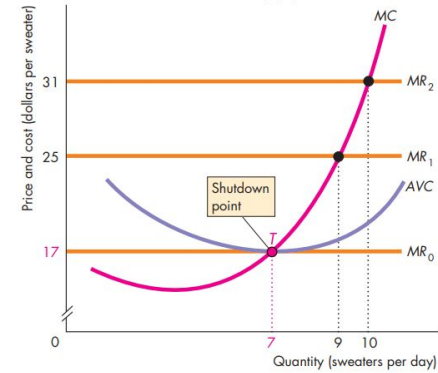
|                                             |             |
|---------------------------------------------|-------------|
| Equilibrium Price (P)                       | = \$17      |
| Quantity of Output produced (Q)             | = 7         |
| Total Revenue (P x Q= \$17 x 7)             | = \$119     |
| Total Cost ( ATC x Q= \$20.14 x 7)          | = \$ 140.98 |
| Total variable Cost (AVC x Q= \$17 x 7)     | = \$119     |
| Total Fixed Cost (TC-TVC= \$ 140.98- \$119) | = \$ 21.98  |
| Profits(TR-TC= \$119- \$ 140.98)            | = - 21.98   |

### Economic Loss= Fixed Cost

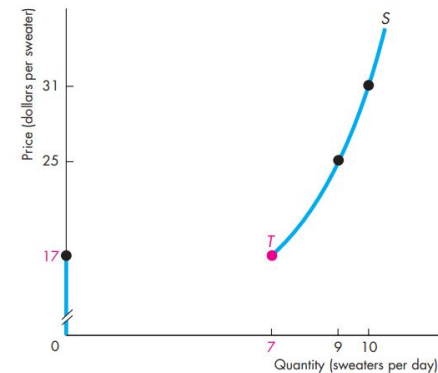
- **Shutdown Decision:** At this point, the firm is **indifferent between operating and shutting down.**

# The Competitive Firm's Short-Run Supply Curve

- The perfectly competitive firm produces (supplies output) in the short run if price is **above** average variable cost.
- It shuts down (does not supply output) if price is below average variable cost.
- Therefore, the **short-run (firm) supply curve is the portion of the firm's marginal cost curve that lies above the average variable cost curve.**



(a) Marginal cost and average variable cost



(b) Campus Sweaters' short-run supply curve



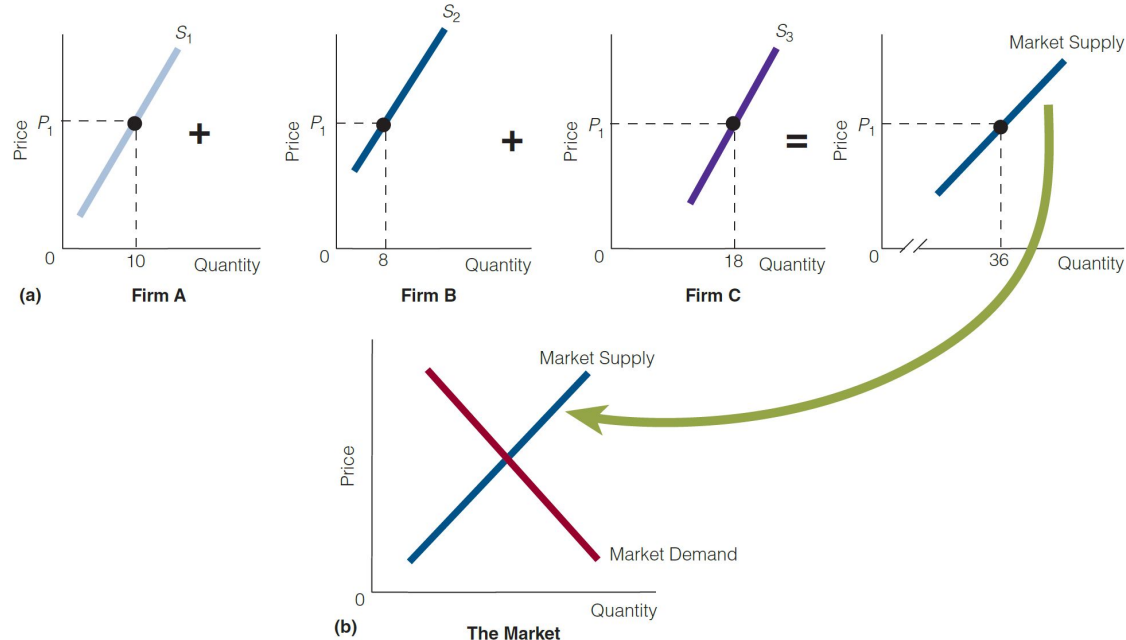
# Short-Run Market Supply Curve

**Short-Run Market Supply Curve:** The horizontal sum of all existing firms' short-run supply curves.

## Deriving the Market (Industry) Supply Curve for a Perfectly Competitive Market

In (a), we add (horizontally) the quantity supplied by each firm to derive the market supply curve. The market supply curve and

the market demand curve are shown in (b). Together, they determine equilibrium price and quantity.



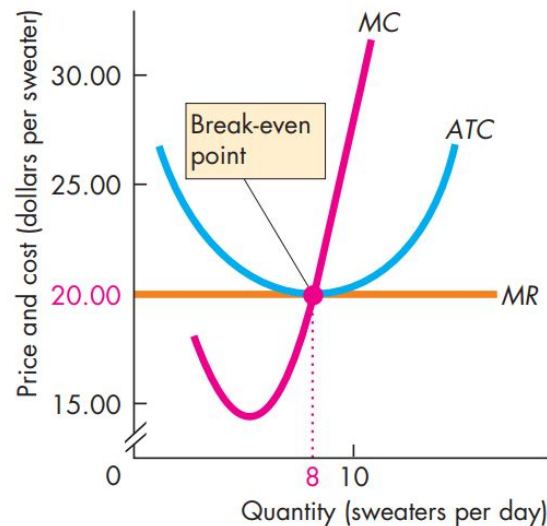
# Profits and Losses in the Short Run

**Three Possible Short-Run Outcomes:** In the short run, the firm might **break even** (make zero economic profit), make an **economic profit**, or incur an **economic loss**.

## 1. Break Even

- From the figure, we observe that the **equilibrium price** of a sweater is **\$20**, and the firm produces **8 sweaters per day**.
- At this output level, the **average total cost (ATC)** is also **\$20** per sweater.
- Since **price equals ATC ( $P = ATC$ )**, the firm is **breaking even**, meaning it earns **zero economic profit**.

Thus, at the **profit-maximizing output**, the firm neither makes a profit nor incurs a loss—it **breaks even**.



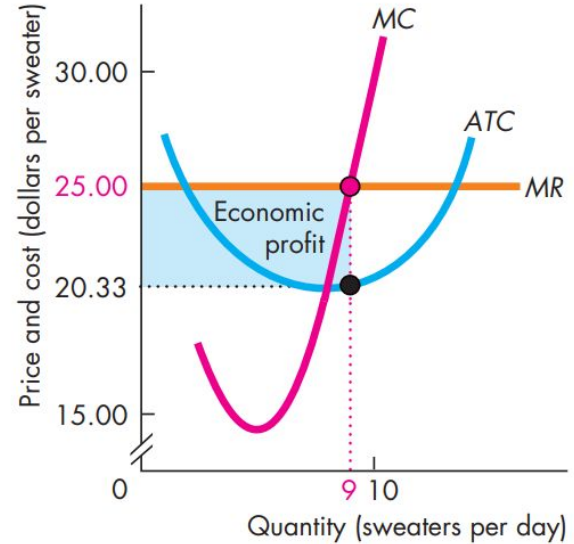
(a) Break even

## 2. Economic Profit

- From the figure, we observe that the **equilibrium price** of a sweater is **\$25**, and the firm produces **9 sweaters per day**.
- At this point, the **price is higher than the average total cost (ATC)**, so the firm earns an **economic profit**.
- The firm makes **\$4.67 profit per sweater** ( $\$25.00 - \$20.33$ ).
- So, total profit is **\$42 a day** ( $\$4.67 \times 9$  sweaters).

In the figure, this profit is shown by a **blue rectangle**:

- The **height** is the **profit per sweater** (\$4.67).
- The **width** is the **number of sweaters** (9).
- The **area** of the rectangle shows the **total profit** (\$42 per day).



(b) Economic profit

### 3. Economic Loss

From the figure, we observe that the **equilibrium price** of a sweater is **\$17**, which is **less than the average total cost (ATC)**.

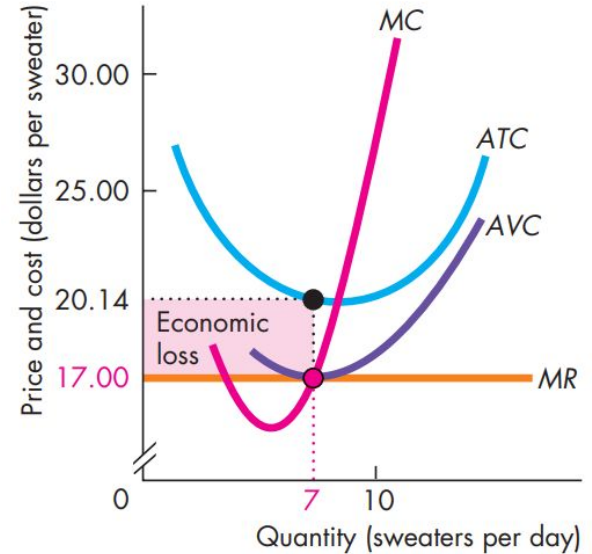
This means the firm is making an **economic loss**.

- The firm produces **7 sweaters per day**
- **Total revenue** is **\$119** ( $7 \text{ sweaters} \times \$17$ ).
- **ATC** is **\$20.14**, so the firm **loses \$3.14 per sweater**.
- Total loss is **\$22 a day** ( $\$3.14 \times 7 \text{ sweaters}$ ).

This loss is shown in the figure as a **red rectangle**:

- The **height** is the **loss per sweater** (\$3.14).
- The **width** is the **number of sweaters** (7).
- The **area** of the rectangle shows the **total loss** (\$22 a day).

If the **price falls below \$17**, the firm will **shut down temporarily** and only lose its **fixed costs**.



(c) Economic loss

# Entry and Exit in the Long-Run

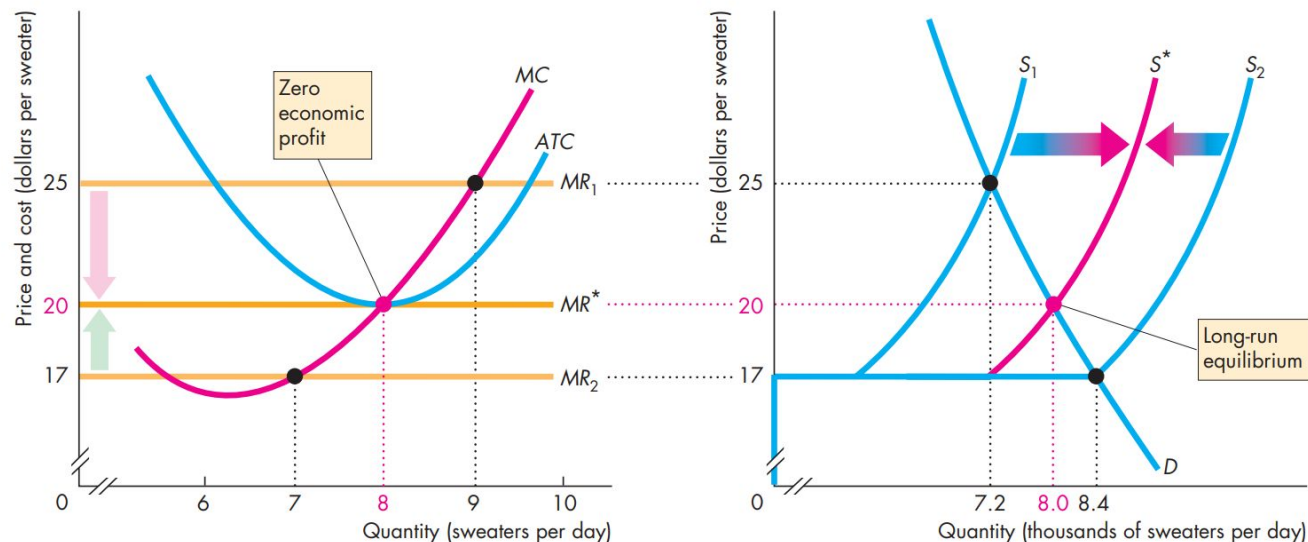
## Firm Entry

- **Entry** occurs in a market when new firms come into the market and the number of firms **increases**.
- Firms **enter a market** when existing firms are earning **economic profit**.
- **New entries increase market supply**, which shifts the **market supply curve to the right**.
- As supply rises, the **market price falls**, which **reduces economic profit**.
- **Entry continues** until **economic profit becomes zero**, at which point **no more firms enter**.

## Firm Exit

- **Exit** occurs when existing firms leave a market and the number of firms **decreases**.
- Firms **exit a market** when they are facing **economic losses**.
- **Exits reduce market supply**, shifting the **market supply curve to the left**.
- As supply decreases, the **market price rises**, which helps **reduce losses**.
- **Exit continues** until **economic loss disappears**, and firms are **no longer leaving** the market.

**FIGURE 12.9** Entry, Exit, and Long-Run Equilibrium



**(a) Campus Sweaters**

Each firm has cost curves like those of Campus Sweaters in part (a). The market demand curve is  $D$  in part (b).

When the market supply curve in part (b) is  $S_1$ , the price is \$25 a sweater. In part (a), each firm produces 9 sweaters a day and makes an economic profit. Profit triggers the entry of new firms and as new firms enter, the market supply curve shifts rightward, from  $S_1$  toward  $S^*$ . The price falls from \$25 to \$20 a sweater, and the quantity produced increases from 7,200 to 8,000 sweaters. Each firm's output decreases to 8

**(b) The sweater market**

sweaters a day and economic profit falls to zero.

When the market supply curve is  $S_2$ , the price is \$17 a sweater. In part (a), each firm produces 7 sweaters a day and incurs an economic loss. Loss triggers exit and as firms exit, the market supply curve shifts leftward, from  $S_2$  toward  $S^*$ . The price rises from \$17 to \$20 a sweater, and the quantity produced decreases from 8,400 to 8,000 sweaters. Each firm's output increases from 7 to 8 sweaters a day and economic profit rises to zero.

# To Summarize

- New firms enter a market in which existing firms are making an economic profit.
- As new firms enter a market, the market price falls and the economic profit of each firm decreases.
- Firms exit a market in which they are incurring an economic loss.
- As firms leave a market, the market price rises and the economic loss incurred by the remaining firms decreases.
- Entry and exit stop when firms make zero economic profit.
- When economic profit and economic loss have been eliminated and entry and exit have stopped, a competitive market is in **long-run equilibrium**.